TASK 2 REPORT: EMBEDDED SYSTEMS LAB SETUP & SENSOR INTERFACING

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1. Objective & Overview

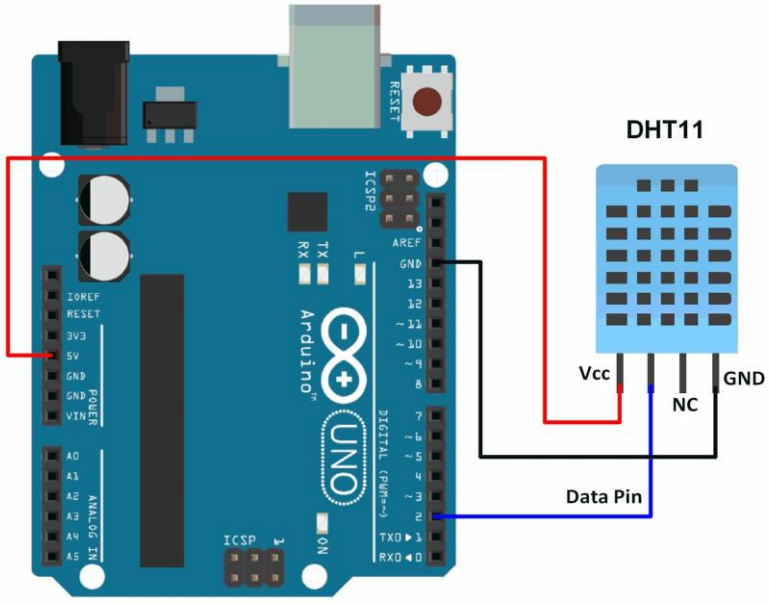
The objective of this project is to interface a **DHT11 Temperature Sensor** with an **Arduino UNO** to measure ambient temperature and generate an alert whenever the temperature exceeds a predefined threshold (30°C).
Monitoring and High Temperature Alert System using DHT11 Sensor and Arduino UNO

2. Development Environment & Hardware Selection

For this task, a hybrid simulation approach using Tinkercad Circuits was selected to verify circuit connections and timing before physical deployment.

- **Microcontroller:** Arduino UNO R3 (ATmega328P)
- **Sensor: DHT11 Temperature Sensor**
- **Actuator:** Red Light Emitting Diode (LED)
- **Passive Components:** 2.2 kΩ Current-Limiting Resistor
- **Software Tool:** Tinkercad Circuits / Arduino IDE (C/C++ Compiler)

3. Circuit Diagram & Pin Mapping



The circuit connects the analog output pin of the TMP36 sensor to pin A0 of the Arduino UNO, while driving the alert LED via digital pin 13.

DHT11 Pin	Arduino UNO Pin
VCC	5V
GND	GND
DATA	Digital Pin 2

4. Working Principle

The DHT11 sensor measures the surrounding temperature and sends the data digitally to the Arduino. The Arduino continuously reads the temperature value and displays it on the Serial Monitor. When the measured temperature becomes greater than **30°C**, the system displays:

"⚠ High Temperature Alert!"

This simple project demonstrates sensor interfacing, data acquisition, and condition-based automation

5. Code Explanation & Logic Breakdown

```
#include "DHT.h"
#define DHTPIN 2
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
void setup() {
  Serial.begin(9600);
  dht.begin();
}
void loop() {
  float temp = dht.readTemperature();
  Serial.print("Temperature: ");
  Serial.println(temp);
  if(temp > 30) {
    Serial.println("⚠ High Temperature Alert!");
  }
  delay(2000);
}
```

6. Output

Normal Condition

Temperature: 27.00

Temperature: 28.00

Temperature: 29.00

High Temperature Condition

Temperature: 31.00

⚠ High Temperature Alert!

Temperature: 32.00

⚠ High Temperature Alert!

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7. Results

- Successfully interfaced the DHT11 temperature sensor with Arduino UNO.

- Temperature data was continuously monitored through the Serial Monitor.
- Alert message was generated whenever the temperature exceeded 30°C.
- The project demonstrated real-time sensor data processing and basic automation logic.

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8. Applications

- Smart Home Automation
- Weather Monitoring Systems
- Industrial Temperature Monitoring
- Greenhouse Environmental Control
- Server Room Monitoring

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9. Conclusion

The DHT11 Temperature Monitoring System was successfully implemented using Arduino UNO. The sensor accurately measured ambient temperature and transmitted the data to the microcontroller. The programmed threshold mechanism effectively generated a high-temperature warning whenever the temperature exceeded 30°C. This project provided hands-on experience in sensor interfacing, Arduino programming, and real-time monitoring systems.